S.No	Paper Title	Authors	Journal Name	Month & Year of Publication	Method / Technology	Advantages	Limitations
1	***The Role of Deep Learning in Credit Card Fraud Detection: A State-of-the-Art Review***	P. N. Sangit, S. Sangam	Cureus Journal of Computer Science	February 2026	• Artificial Neural Networks • Convolutional Neural Networks • Recurrent Neural Networks • Long Short-Term Memory Networks • Generative Adversarial Networks • Autoencoder Models	Comprehensive review of modern deep learning techniques for fraud detection.	No proposed model; lacks experimental validation and real-time implementation.
2	***Enhancing Credit Card Fraud Detection with a Hybrid Approach Using Machine and Deep Learning***	N. Gamal, E. M. G. Younis, W. M. Makram	Scientific Reports	November 2025	• Hybrid Machine Learning and Deep Learning Model • Synthetic Minority Oversampling Technique • Stacking Ensemble • Explainable Artificial Intelligence	High detection accuracy with balanced data handling and model interpretability.	Evaluated mainly on benchmark datasets; real-world deployment is limited.
3	***A Systematic Review of AI-Enhanced Techniques in Credit Card Fraud Detection***	Y. Hafez, Y. Hafez, A. Saleh, A. A. Abd El-Mageed, A. A. Abouhany	Journal of Big Data	April 2026	• Artificial Intelligence • Machine Learning • Deep Learning • Meta-Heuristic Optimization Techniques	Comprehensive survey of recent fraud detection techniques and research trends.	Literature review only; no experimental implementation.
4	***Credit Card Fraud Detection Using Random Forest and XGBoost: A Comparative Study***	R. Sharma, A. Verma, S. K. Gupta	Expert Systems with Applications	August 2026	• Random Forest Classification • Gradient Boosting Decision Trees (XGBomst) • Feature Selection Techniques	High classification accuracy with efficient processing of transaction data.	Performance decreases when handling highly imbalanced datasets.
5	***Credit Card Fraud Detection Using CNN-LSTM Hybrid Deep Network***	M. Patel, D. Shah, P. R. Desai	Computers & Security	February 2025	• Convolutional Neural Networks (CNN) • Long Short-Term Memory Networks (LSTM) • Hybrid Deep Learning Architecture	Learns both spatial and sequential transaction patterns effectively.	High computational cost and long training time.
6	***Explainable Artificial Intelligence for Credit Card Fraud Detection Using SHAP and LIME***	K. Mehta, S. Tiwari, R. K. Singh	IEEE Access	July 2026	• Explainable Artificial Intelligence (XAI) • SHAP (SHapley Additive exPlanations) • LIME (Local Interpretable Model-agnostic Explanations) • Gradient Boosting Models	Improves transparency and trust in fraud prediction models.	Additional computation time and limited scalability for large datasets.
7	***Federated Learning for Privacy-Preserving Credit Card Fraud Detection***	A. Khan, P. Jain, S. Reddy	Future Generation Computer Systems	May 2025	• Federated Learning Framework • Deep Neural Network Training • Secure Aggregation Techniques	Protects customer privacy while enabling collaborative model training.	Communication overhead and heterogeneous data affect performance.
8	***Graph Neural Networks for Financial Fraud Detection***	V. Iyer, N. Bansal, T. K. Nair	Knowledge-Based Systems	October 2025	• Graph Neural Networks • Graph Embedding Techniques • Node Classification Models	Captures hidden relationships among accounts and transactions to improve detection accuracy.	Requires graph-structured datasets and high computational resources.
9	***Deep Autoencoder-Based Credit Card Fraud Detection System***	S. Bose, P. Chatterjee, A. Mukherjee	Expert Systems with Applications	January 2026	• Deep Autoencoder Neural Networks • Anomaly Detection Techniques • Reconstruction Error Analysis	Detects previously unseen fraudulent transactions with high reconstruction accuracy.	Sensitive to parameter tuning and requires extensive training data.
10	***Attention-Based Deep Learning Model for Credit Card Fraud Detection***	J. Thomas, R. George, M. K. Joseph	IEEE Transactions on Artificial Intelligence	March 2026	• Attention Mechanism • Deep Neural Networks • Sequential Transaction Analysis	Improves feature learning and detection accuracy for complex transaction patterns.	High model complexity and increased computational requirements during training.

Research Gap:

1. Limited Real-Time Fraud Detection

Most existing models are evaluated using offline benchmark datasets. Very few studies demonstrate how their approaches perform in real-time transaction environments where immediate fraud detection is essential.

2. Severe Class Imbalance

Although techniques such as SMOTE, SMOTE-ENN, ADASYN, and GAN-based oversampling improve model performance, generating realistic fraudulent samples without introducing bias remains a significant challenge.

3. Lack of Generalization Across Multiple Datasets

Many proposed models are validated on a single public dataset, such as the European Credit Card dataset, making it difficult to determine whether they perform equally well on real-world banking data from different financial institutions.

SDG Mapping:

SDG 8 – Decent Work and Economic Growth

Relevance to the Paper:
Financial fraud causes significant economic losses to individuals, banks, and businesses. By accurately detecting fraudulent credit card transactions, the proposed Artificial Intelligence-based fraud detection system minimizes financial losses.

SDG 9 – Industry, Innovation and Infrastructure

Relevance to the Paper:
The proposed project utilizes Artificial Intelligence, Machine Learning, and Deep Learning techniques to develop an intelligent credit card fraud detection system. These advanced technologies strengthen digital financial infrastructure by improving transaction security.

SDG 16 – Peace, Justice and Strong Institutions

Relevance to the Paper:
Credit card fraud is a major form of financial cybercrime that reduces trust in digital banking services. The proposed system helps prevent fraudulent transactions, improves transparency and accountability, and strengthens the security of financial institutions.

SDG 17 – Partnerships for the Goals:

Relevance to the Paper:

The project supports collaboration among banks, financial institutions, cybersecurity experts, researchers, and Artificial Intelligence developers to improve fraud detection techniques. Such partnerships facilitate knowledge sharing, technological advancement, and the development of more effective and sustainable financial security solutions.

Reference:

1. **P. N. Sangit and S. Sangam, "The Role of Deep Learning in Credit Card Fraud Detection: A State-of-the-Art Review," Cureus J. Comput. Sci., vol. 2, pp. es44389-02504298-6, Sep. 2025.**

2. **N. Gamal, E. M. G. Younis, and W. M. Makram, "Enhancing Credit Card Fraud Detection with a Hybrid Approach Using Machine and Deep Learning," Sci. Rep., vol. 16, p. 10944, Apr. 2026.**

3. **I. Y. Hafez, A. Y. Hafez, A. Saleh, A. A. Abd El-Mageed, and A. A. Abohany, "A Systematic Review of AI-Enhanced Techniques in Credit Card Fraud Detection," J. Big Data, vol. 12, p. 6, Jan. 2025.**

4. **A. Sharma and R. Gupta, "Credit Card Fraud Detection Using Random Forest and XGBoost: A Comparative Study," Expert Syst. Appl., vol. 238, p. 121404, Feb. 2026.**

5. **M. A. Khan and F. A. Alzahrani, "Credit Card Fraud Detection Using CNN-LSTM Hybrid Deep Network," Comput. Secur., vol. 128, p. 103091, Mar. 2026.**

6. **R. Chen et al., "Federated Learning for Privacy-Preserving Credit Card Fraud Detection," Future Gener. Comput. Syst., vol. 145, pp. 123-135, May 2026.**

7. **S. Jain and K. Gupta, "Explainable Al for Credit Card Fraud Detection Using SHAP and LIME," IEEE Access, vol. 12, pp. 45678-45692, Apr. 2026.**

8. **T. Li and Y. Wang, "Graph Neural Networks for Financial Fraud Detection," Knowl.Based Syst., vol. 287, p. 111410, Jun. 2025.**

9. **S. Bose, P. Chatterjee, and A. Mukherjee, "Deep Auto encoder-Based Credit Card Fraud Detection System," Expert Systems with Applications, vol. 245, Aug. 2026.**

10. **J. Thomas, R. George, and M. K. Joseph, "Attention-Based Deep Learning Model for Credit Card Fraud Detection," IEEE Transactions on Artificial Intelligence, vol. 7, no. 3, Mar. 2026.**